Juejue Wang

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Research Interests

- Causal inference, sensitivity analysis, quasi-experimental designs
- Functional data analysis, neuroimaging

Education

University of Washington Seattle, WA
Ph.D. in Statistics, GPA: 3.87/4.00 09/2022 – 08/2027

University of Michigan Ann Arbor, MI
M.S. in Applied Statistics (1 year Accelerated), GPA: 4.00/4.00 04/2021 – 04/2022

University of Michigan Ann Arbor, MI
B.S. in Statistics (Honors), GPA: 4.00/4.00 09/2018 – 04/2021
B.S. in Economics (Honors), GPA: 3.96/4.00

Research Experience

University of Washington, Department of Statistics Seattle, WA
Research Assistant in Sensitivity Analysis | Advisor: Carlos Cinelli 03/2025 – Present

- Derived a novel characterization of the omitted-variable bias for the ATT in canonical DiD designs, improving the communication of sensitivity results in applied work.
- Provided Neyman-orthogonal scores for DiD, allowing valid machine-learning-based estimation and inference for nuisance parameters.

Research Assistant in Functional Data Analysis | Advisor: Eardi Lila 06/2024 – Present

- Developed an R package for geometry-aware discriminant analysis of functional data on nonlinear manifold domains, based on Lila et al.'s Functions on Surfaces (FoSs) framework.
- Extended the framework to multi-class classification and multi-dimensional functional predictors, preserving geometry-aware spatial regularization and covariance-free estimation.

University of Michigan, Department of Statistics Ann Arbor, MI
Independent research in Topic modeling | Advisor: Johann Gagnon-Bartsch 09/2020 – 05/2022

- Evaluated five document co-clustering algorithms on Twitter data, using Latent Dirichlet allocation (LDA) as a topic-modeling baseline (Python, R).
- Applied single-cell RNA-seq data clustering methods (Seurat and Monocle3) for Twitter data and developed a semi-automated pipeline that achieves accuracy comparable to hand-created classification while remaining efficient for many latent topics.

Independent research in Mechanism design | Advisor: Tilman Börgers 09/2020 – 04/2021

- Analyzed limitations of widely used medical-supply priority systems using mathematical modeling.
- Applied static game theory to show how a proposed reserve system reduces qualification overstatement and promotes stakeholder compromise.
- Developed policy recommendations for COVID-19 vaccine distribution based on these findings.

University of Michigan, Big Data Summer Institute Ann Arbor, MI
Summer research in Survival analysis [code] | Advisor: Xiang Zhou 06/2019 – 07/2019

- Applied k-means/hierarchical clustering and survival analysis against over 300 million entries of cancer tissue data and 11 existing deconvolution methods to predict the survival time, using R, Python, and Linux Cluster.
- Presented results to UM Symposium on Big Data, Human Health, and Statistics, and the 2019 Northern California Computational Biology Student Symposium.

Papers

Juejue Wang, Pedro Sant'Anna, Victor Chernozhukov and Carlos Cinelli (working paper, 2026). "Omitted Variable Bias in Difference-in-Differences Designs". [draft]

Juejue Wang, Julian Diefenbacher, Jannis Kueck, Martin Spindler, Victor Chernozhukov, and Carlos Cinelli (working paper, 2026). "Sensitivity Analysis for (Local) Average Treatment Effects in Instrumental Variable Models".

Juejue Wang (2021). "Comparison of Document Co-clustering algorithms and Application of Single-cell RNA-seq Data Clustering to Twitter Data". (Undergraduate honors thesis in statistics, University of Michigan) [link][code]

Juejue Wang (2021). "Examination of the Reserve System for Medical Resource Allocation in Pandemics". (Undergraduate honors thesis in economics, University of Michigan) [link]

Talks

- Joint Statistical Meetings (JSM 2026), August 2026, Boston, Massachusetts (upcoming)
- American Causal Inference Conference (ACIC 2026), May 2026, Salt Lake City, Utah
- European Causal Inference Meeting (EuroCIM 2026), April 2026, Oxford, England
- University of Washington, Data Science Seminar, April 2026, Seattle, Washington
- University of Washington, Causal Reading Group, April 2026, Seattle, Washington
- University of Washington, TGIF Lab, Feb. 2026, Seattle, Washington
- IEEE International Conference on Big Data (Poster), Dec. 2021, Orlando, Florida
- New Directions in Analyzing Text as Data Conference (TADA 2021, Poster), Oct. 2021, Ann Arbor, Michigan
- Northern California Computational Biology Student Symposium, Oct. 2019, Davis, California
- UM Symposium on Big Data, Human Health, and Statistics, July 2017, Ann Arbor, Michigan

Teaching Experience

Department of Statistics, University of Washington Seattle, WA

Teaching Assistant 09/2022 – Present

- STAT 221 (F23, Sp23): Statistical Methods for the Social Sciences
- STAT 220 (W23): Statistical Reasoning
- STAT 390 (W25): Statistical Methods in Engineering and Science
- STAT 394 (Sp24): Probability I
- STAT 424 (Sp25): Generalized Linear Models
- DATA 558 (Sp26): Statistical Machine Learning for Data Scientists
- STAT 581 (F24, F25, F26): Advanced Theory of Statistical Inference I
- STAT 582 (W24, W26): Advanced Theory of Statistical Inference II

Directed Reading Program (DRP) mentor 03/2026 – 06/2026

Mentored an undergraduate student on sensitivity analysis for difference-in-differences designs, covering omitted variable bias and debiased machine learning methods applied to real economic datasets.

Department of Statistics, University of Michigan Ann Arbor, MI

Graduate Student Instructor 08/2021 – 12/2021

- STAT 306: Introduction to Statistical Computing (in R)

Co-mentor of Data Mining Group, Big Data Summer Institute (BDSI) 06/2021 – 07/2021

Led technical guidance for data-mining research. Deployed projects using Docker.

Awards

- JSM Early Career Travel Award, 2026
- Birnbaum Student Recruitment Fellowship, University of Washington, 2022
- Sims Honor Scholarship Prize, University of Michigan, 2021
- Highest Honors in Statistics and Economics, University of Michigan, 2021
- James B. Angell Scholar, University of Michigan, 2020 - 2022
- University Honors, University of Michigan, 2018 - 2021